Medoid Trio

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Multimodal Reasoning for Image Sequences

1. 문제 정의

- 연속된 image sequences를 가장 잘 설명하는 text description을 찾기

2. Dataset

- Train: Image sequence + one description
- Test: Image sequence + five description (1 right desc, 4 wrong desc)

Text Augmentation

데이터 증강: 오답 생성

정답: 'A person opens a red bag.'

Method 1. spaCy + WordNet

Rule-based 단어 치환

- **Antonym**: A person **closes** a red bag.
- **Attribute Change**: A person opens a **blue** bag.
- **Object Replacement:** A person opens a red **box**.

Method 2. LLM-based Generation

다양한 유형의 오답 설명 생성

- **Negation**: A person **does not open** a red bag.
- **Action Change**: A person **removes an object** from a red bag.
- **Temporal Change**: A person closes the bag **before** opening it.
- **Attribute Change**: A person opens a **large green** bag.

데이터 증강: 데이터셋 생성

생성 규칙

L1: 한 단어 단위의 변경

L2: 문장 구조 단위의변경

L3: LLM기반 rewrite

L4: 다른 episode의 description

정답 데이터, L1, L2, L3, L4 데이터를 하나 씩 섞어 하나의 Train Set으로 사용하자.

L1, L2 : spaCy + WordNet, Rule-based 단어 치환

정답: ‘A person opens a red bag.’

- **Subject replacement** : A **gorilla** opens a red bag.
- **Verb Change**: A person **runs** a red bag.
- **Object Replacement:** A person opens a red **cap**.

SmolVLM이 문장만 보고 오답을 탐지한 비율 83%

→ 역으로 SmolVLM을 활용한 부자연스러운 문장 탐지를 통해 선별된 오답 생성

Image Augmentation

연구 배경

기존 Vision-Language Model은 영상 내 동작의 방향과 발생 순서를 구분하는 데 한계가 있음.

참고하여 설계

Test of Time: Instilling Video-Language Models with a Sense of Time

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bpiyush.github.io/testoftime-website

.02074v2 [cs.CV] 25 Mar 2023

Abstract

Modelling and understanding time remains a challenge in contemporary video understanding models. With language emerging as a key driver towards powerful generalization, it is imperative for foundational video-language models to have a sense of time. In this paper, we consider a specific aspect of temporal understanding: consistency of time order as elicited by before/after relations. We establish that seven existing video-language models struggle to understand even such simple temporal relations. We then question whether it is feasible to equip these foundational models with temporal awareness without re-training them from scratch. Towards this, we propose a temporal adaptation recipe on top of one such model, VideoCLIP, based on post-pretraining on a small amount of video-text data. We conduct a zero-shot evaluation of the adapted models on six datasets for three downstream tasks which require varying degrees of time awareness. We observe encouraging performance gains especially when the task needs higher time awareness. Our work serves as a first step towards



A. Dog runs away *before* it brings a ball to the man
B. The dog brings a a ball to the man *before* it runs away
C. The baby eats food *after* it looks into the camera
D. The baby looks into the camera *after* it eats food

Figure 1. Can you match the correct video-text pairs? Understanding the time order of events across video and language is necessary to be able to solve this task. See footnote on next page for answers.

[Bagad, P., Tapaswi, M., & Snoek, C. G. M. (2023). Test of time: Instilling video-language models with a sense of time. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 2503-2516). [1]

ViTAL-X: Video-Text Alignment with Cross-Modal Temporal Edits

Sethuraman T V[1,2], Savya Khosla[1], Onkar Kishor Susladkar[1], Aditi Tiwari[1], Seoung Wug Oh[2], Kushal Kafle[2], Joon-Young Lee[2], Derek Hoiem[1], and Simon Jenni[2]

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Abstract. Video-text models adapted from image-text architectures (e.g., CLIP) frequently exhibit *temporal blindness*—the inability to perceive fundamental cues like order, direction, and motion dynamics. Standard datasets mask this limitation by enabling models to exploit static spatial shortcuts. To systematically evaluate this, we introduce **XTE-Bench**, a diagnostic probe revealing that even large-scale video-language models struggle with basic temporal reasoning, indicating that parameter scaling alone is insufficient to resolve this flaw. To address this, we propose **Cross-Modal Temporal Edits (XTE)**, a self-supervised framework that injects precise temporal supervision. By performing synchronized video-text transformations, XTE generates hard temporal negatives without manual annotation. We instantiate this with **ViTAL-X**, a lightweight model that equips frozen image-text backbones with temporal awareness while preserving their foundational spatial knowledge. Across six temporal benchmarks, ViTAL-X achieves state-of-the-art performance. Utilizing only 0.4B parameters and 1M training clips, ViTAL-X outperforms 7B-parameter models and surpasses baselines trained on 600× more data. These results demonstrate that targeted, high-quality temporal alignment provides a highly efficient alternative to pure scaling.

[Sethuraman, T. V., Khosla, S., Susladkar, O. K., Tiwari, A., Oh, S. W., Kafle, K., Lee, J.-Y., Hoiem, D., & Jenni, S. (2026). ViTAL-X: Video-Text Alignment with Cross-Modal Temporal Edits. In Computer Vision – ECCV 2026 (pp. 123-142). Springer Nature Switzerland. https://doi.org/10.1007/978-3-032-37174-4_8

영상 기반 데이터 증강

Hypothesis: Temporal data augmentation improves the model's ability to recognize action direction and event order.

Original video sequence

A	→	B	→	C
Bag closed		Opening		Bag open

1. Temporal reversal

C → B → A

2. Segment shuffling

B → A → C

3. Reversed video + verb flip

Original: **opens** the bag → Edited: **closes** the bag

Illustrative sequences. Actual transformations must preserve the correct visual interpretation; reversing frames does not always produce a valid opposite action.

Temporal Order Contrastive Loss



정답 선지가 주어졌을 때
A->B->C, C->B->A 중 올바른 배열의 이미지를 선택하도록 학습

Model

1. Image-Sequence Model

SigLIP frame features + Temporal Transformer

- SigLIP로 Image Encoding
- temporal Transformer로 Encoded Image를 Video Vector로 변환.
- SigLIP로 만든 Text Embeddig과 비교하여 점수를 매긴다.

SeqAlign

- SigLIP로 Text Encoding 후 Sequence에 맞춰 분할.
- Image Frame들과 비교하여 점수를 매긴다.

Global + Local SigLIP

- Global: 전체 Image의 흐름, Local: 연속된 Image의 연관성.
- 이를 학습에 포함시켜 Sequence 정보를 더 잘 배운다.

1. Image-Sequence Model

Sequence 정보가 전혀 학습되지 않았다.

세 모델의 Public Score → 0.365-0.416

→ Image가 읽히지 않으면 Text를 읽어보자.

2. Text Centrality

1. Word Medoid

- 가설: Test set의 오답은 Ground Truth로부터 단어 단위의 변경으로 생성되었을 것이다.
- 가장 많은 공통 단어를 포함한 문장이 정답일 것이다.

2. Meaning Medoid

- 가설: Test set의 오답은 Ground Truth로부터 의미 단위의 변경으로 생성되었을 것이다.
- Text Encoding 후 Cosine Similarity를 비교하여 중심에 있는 문장이 정답일 것이다.

3. Additional Idea

- Word Medoid와 Meaning Medoid 결합
- 3-5 gram cosine을 이용하여 조금 더 긴 흐름 파악

3. Zero-shot Baseline: VLM Likelihood Ranking

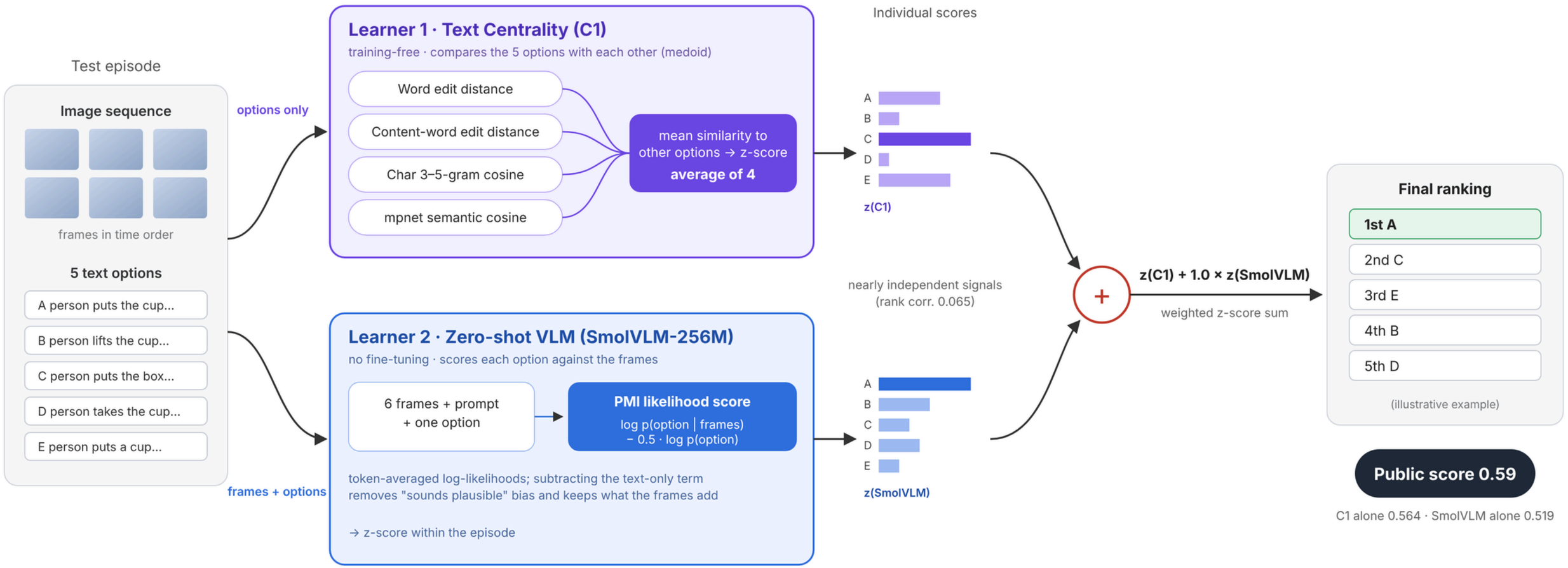
학습 없이 사전학습된 SmolVLM-256M-Instruct를 그대로 사용
각 선지 문장이 "영상을 보고 생성될 확률"로 순위를 매김

Finetuning 진행시 정확도가 떨어지는 모습 확인

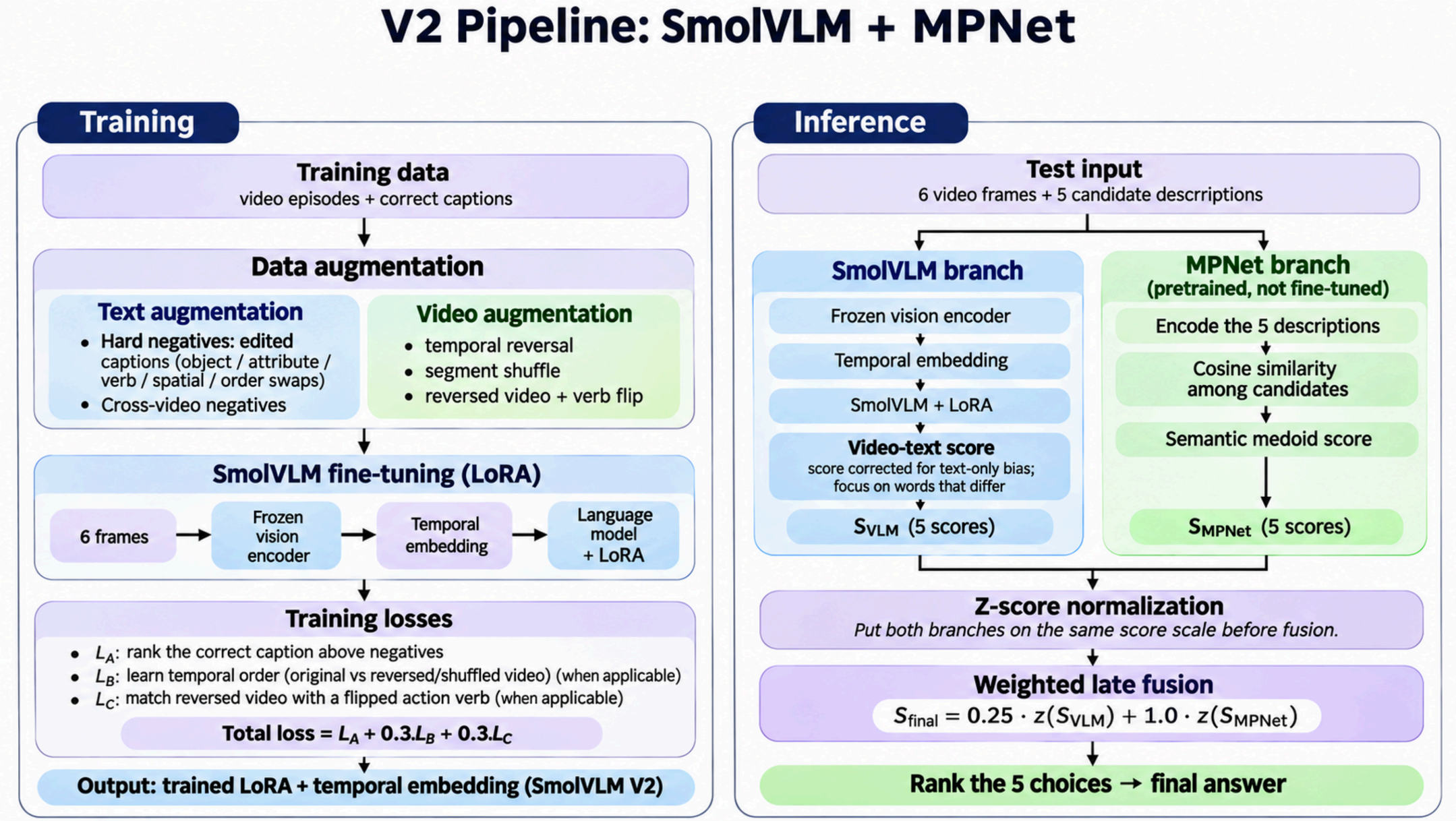
1. LLM이 작성하는 문장의 형태를 암기
2. 부족한 데이터셋을 사용하여 학습시 기존의 정확도에 악영향

3. Ensemble Learning

Ensemble: Text Centrality (C1) + Zero-shot SmolVLM



4. Best Fine Tuned Model: Architecture & Pipeline



Loss

$$\mathcal{L}_A = -\log \frac{\exp(S(s^+, V))}{\sum_{s \in T_A} \exp(S(s, V))}$$

1. Loss A — Listwise Ranking Loss
Rank the correct caption higher

$$\mathcal{L}_B = -\log \frac{\exp(M(s^+, V))}{\exp(M(s^+, V)) + \exp(M(s^+, V_{\text{rev}})) + \exp(M(s^+, V_{\text{swap}}))}$$

2. Loss B — Temporal Order Contrastive Loss
Prefer the original frame sequence over reversed and shuffled sequences.

$$\mathcal{L}_C = -\log \frac{\exp(S(\text{flip}(s^+), V_{\text{rev}}))}{\exp(S(\text{flip}(s^+), V_{\text{rev}})) + \exp(S(s^+, V_{\text{rev}}))}$$

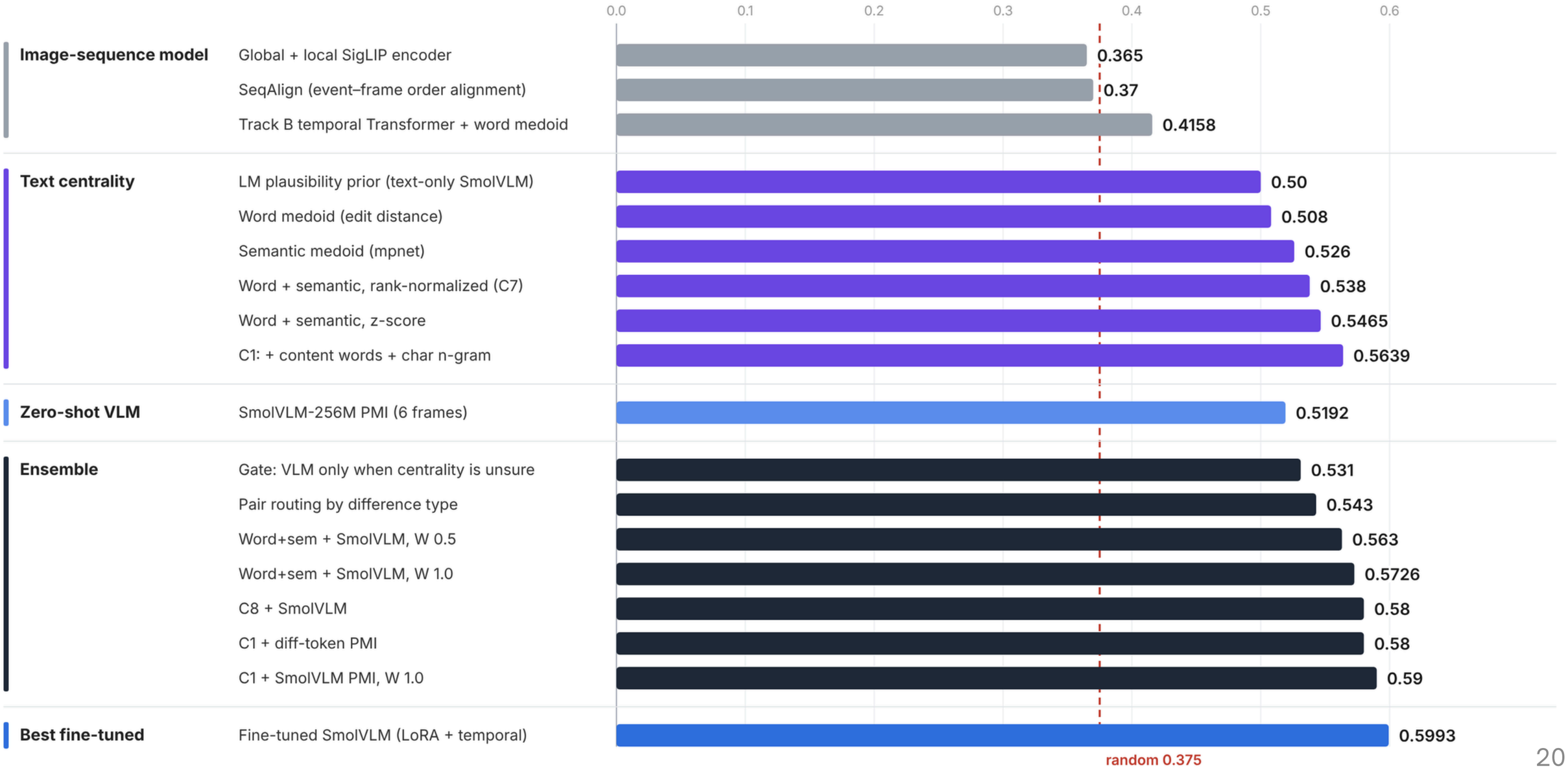
3. Loss C — Action-Direction Consistency Loss
Prefer the flipped action caption over the original caption when evaluating the reversed video.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_A + 0.3\,1_B\mathcal{L}_B + 0.3\,1_C\mathcal{L}_C$$

Total Training Loss

Public Score by Model

Grouped by approach, in the order we explored them



InThon 2026 public leaderboard. Diff-token PMI and C8 blends use SmolVLM zero-shot as the visual signal.

Conclusion & Future Work

가장 강한 단서는 Text centrality였다. -> Training-free text centrality(**medoid**)만으로 0.564달성
Zero-shot VLM(SmolVLM PMI)이 거의 독립적인 visual signal을 더해 주었다(rank corr. 0.065, 단독 0.519)
Text centrality와 Zero-shot 두 신호의 단순 z-score ensemble로 0.59,
간단한 finetuning 후 최종 모델 0.599를 달성하였다.

Fine-grained temporal reasoning: 동사 방향, event order, left/right만 다른 선지는 여전히 풀지 못한다.
Test distribution에 맞는 hard negative 구축이 이루어졌는지 점검이 부족했다.
고정 weight 대신 learned / confidence-aware ensemble을 통해 더욱 향상시킬 수 있을 것 같다.